

Advanced Theory — Module 04: Cognition — Study Questions

1. What is the problem of temporal depth? Why can't a flat model handle multiple timescales?
2. Describe the architecture of a deep temporal model. How do levels differ in their "clock speed"?
3. How does the transition model at level l incorporate contextual influence from level $l+1$?
4. Why does only the lowest level generate observations? What do higher levels "observe"?
5. What are ascending messages? What information do they carry and in what form?
6. What are descending messages? How do they constrain lower-level inference?
7. How does precision weighting determine the balance between bottom-up and top-down messages?
8. How does multi-scale planning work? Give a concrete example with at least 3 levels.
9. How is expected free energy computed at each level?
10. How do higher-level policies constrain lower-level policies?
11. What is generative replay? How does it implement mental simulation?
12. How does counterfactual reasoning work mathematically? How are policies compared?
13. What computational role does dreaming play in deep temporal models?
14. How does the deep temporal model relate to the cortical hierarchy proposed in neuroscience?
15. What is the connection between temporal depth and consciousness?
16. How does the model handle unexpected events that violate high-level expectations?
17. What happens when ascending and descending messages are highly discrepant?
18. How does learning differ across levels (fast at lower levels, slow at higher levels)?
19. How does the model explain the "binding problem" — integrating information across timescales?
20. Critically evaluate: Does adding temporal depth always improve inference, or does it have computational costs?